



Multi-criteria itinerary planning for the mobile sink in heterogeneous wireless sensor networks

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Abstract

The mobile sink is usually applied to improving the performance of wireless sensor networks (WSNs). This improvement is subject to the itinerary plan, through which the sojourn times and settlement locations are determined. This paper simultaneously considers multiple criteria like data merit, latency, deadline, used memory, and travel time, then presents a multi-objective Integer Quadratically Constrained Quadratic Programming (IQCQP) for planning itinerary of mobile sink in WSNs. In practice, this programming should be solved in a centralized node and it is computationally NP-hard, but the mathematical formulation can help estimate the problem characteristic and figure out the optimal path for the mobile sink. In addition, this paper proposes a multi-attribute algorithm where the sink gathers attributes of possible sites and compares them by applying fuzzy Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). Therefore, the neighboring adjacent sites are ranked by TOPSIS, then the best site is selected and the mobile sink moves there. The proposed algorithm has lower complexity and utilizes local information to make a decision, therefore it is can handle dynamic features of WSNs. The proposed methods are assessed through rigorous simulations, the results reveal that the proposed distributed algorithm outperforms counterparts in terms of performance metrics (total merit, latency, gathered, expired, and lost data) in a heterogeneous WSN.

Keywords Heterogeneous wireless sensor networks · Multi criteria decision making · Mobile sink · Fuzzy TOPSIS

1 Introduction

Wireless sensor networks (WSNs) have been applied in monitoring the environment. In terms of both computation and energy capacity, these networks are characterized by low resources. In a WSN, the sensors that are located close to the sink have to participate in relaying data on behalf of other sensors, which leads to depleting their energy very quickly (Kaur and Sood 2015; Zhu et al. 2018).

The exploitation of sink mobility in improving the efficiency of a WSN has become a major issue of interest among the researchers, because it provides some advantages like the ones introduced in Dong et al. (2014), Kaswan et al. (2017, 2018), Alsaafin et al. (2018) involve: (1) both energy consumption and interference are reduced because the nodes need small transmission power to communicate with the

sink; (2) energy consumption is balanced, so the energy-holes are reduced and the network lifetime is prolonged; (3) sparse and disconnected networks can be handled better; (4) the operational cost of WSNs is reduced because fewer sensors are required to collect data from an environment and obstacles that exist in the monitored area (where the sensors can not communicate with each other) can be bypassed.

If the sink's mobility is controllable, higher efficiency is assured, however, its throughput depends on the mobile sink travel path. The itinerary plan (named trajectory, path selection, travel path, and path design as well) provides the site's sequence (named settlement or pre-arranged locations, parking position, and rendezvous points as well) and the sojourn times for traveling mobile sink. In planning an itinerary, different objectives like: sensors' density (Kaswan et al. 2017), energy consumption (Alsaafin et al. 2018), the data merit or value of information (Gjanci et al. 2018), deadline (Tashtarian et al. 2015), latency or delay (Alhasanat et al. 2015; Ji 2019; Mehto et al. 2020a), coverage (Wang et al. 2020), and energy density (Ji 2019; Nitesh et al. 2019) are of concern. Researchers have considered these objectives

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and introduced many single and multi-objective optimization problems for planning an itinerary in WSNs (Kaswan et al. 2018). There are some essential and well-known drawbacks relating to optimal itinerary planning in practical WSNs (e.g. its computational complexity is NP-Hard and the considerable energy cost of gathering attributes' volumes). Therefore, it can be concluded that this optimization problem is not beneficial in dynamic and heterogeneous WSNs. To apply optimization models in practical WSNs, some studies assume that the sink rarely moves (Kaswan et al. 2017, 2018; Khalily-Dermany and Nadjafi-Arani 2017). However, if the sensors store the sensed data until the sink meets their sites, then latency increases considerably. On the contrary, some studies have made an unrealistic assumption that all the sensors are homogeneous (e.g. having a constant rate of traffic generation). In this situation, the attributes' volumes are predictable overtime, however, the practical WSNs may generate dynamic and heterogeneous traffic (Din et al. 2020).

Distributed and centralized methods can be adopted in planning an itinerary for the mobile sink. The distributed algorithms (the sink applies local information and draws the moving decisions autonomously) can better adapt to different WSNs' conditions because they can handle dynamic features, are robust to the sensors' faults, and have low complexity (Alsaafin et al. 2018; Yun et al. 2013). In providing a distributed approach for planning an itinerary, some greedy algorithms have been introduced, which make the locally optimal choice at each stage. They greedily move the sink towards a location that has a higher volume for a considered attribute or a higher weighted value (Salarian et al. 2014). For instance, the mobile sink is moved to a specific area where there exist more source nodes (Zhu et al. 2016), higher remaining energy (Tan et al. 2009), etc. However, greedy algorithms do not know what lies ahead of the current greedy selection, so they may fail to find the globally optimal solution. To the best knowledge of the authors, in all the available distributed algorithms in this context, at the most, three objects are of concern, and no algorithm considers all application requirements of heterogeneous WSNs.

This paper considers itinerary planning in WSNs as a multi-criteria problem and supports various types of WSN applications that require efficiency in different attributes including the data merit, deadline, used memory, latency, travel time, density, and fairness. In the first step, a multi-objective optimization problem is proposed which should be executed in a central node to obtain an itinerary plan. It is a multi-objective Integer Quadratically Constrained Quadratic Programming (IQCQP). Many algorithms can be applied in solving this optimization model, but IQCQPs are computationally NP-hard and cannot be solved in a polynomial time. In addition, it needs to gather attributes' volumes into a central node and then provide the optimal solution. If the network's parameters change, then this process (gathering

and solving) should be repeated every time that the sink wants to travel. Therefore, it undergoes a noticeable cost for gathering the attributes' volumes and solving the emerged optimization problem, but, it can help estimate the characteristics and properties of the hardness of itinerary planning. However, the ideal itinerary plan determined by solving this optimization problem helps assess the itinerary plans. Then, a multi-attribute algorithm for planning the itinerary of the mobile sink in heterogeneous WSNs is proposed. This algorithm is distributive because it collects the sink's vicinity information to decide where to move and when. It considers the sink movement as a Multi-Attributes Decision Making (MADM) problem, and the fuzzy Technique of Order Preference Similarity to the Ideal Solution (TOPSIS) algorithm is applied to propose an itinerary plan. To decide which one of the potential future sites is more appropriate, the sink gathers some information on the considered attributes of possible sites and compares them through the decision-making system. More clearly, the neighboring adjacent sites are ranked by TOPSIS, and then the best site is selected. Applying fuzzy TOPSIS in obtaining a distributed algorithm for planning the itinerary of the mobile sink with multiple attributes is the main contribution of this article which paves the way for more studies to be employed in the heterogeneous WSNs field. The efficiency of this algorithm is evaluated and compared in comparison with the optimization model and its counterparts. The simulation results indicate that the proposed algorithm can effectively specify an itinerary plan for the mobile sink in dynamic and heterogeneous WSNs.

This article is organized as follows: the literature is reviewed in Sect. 2; the system model, preliminaries, and problem statements are presented in Sect. 3; the proposed algorithm is introduced in Sect. 4; the performance evaluation is presented in Sect. 5 and the article is concluded in Sect. 6.

2 Literature review

In most WSNs, the sensors collect information and report to a static sink, cooperatively. The static sink is unable to operate efficiently in most scenarios because it shifts the burden from the sensors located far from the sink, to those in the vicinity of the sink. The sink mobility assumption may be especially useful where it moves inside the environment to collect data from the sensors. Researchers have introduced three classes for sink mobility: random, predetermined, and controlled.

The random sink mobility is very simple to implement where the casual motion of mobile entities, like humans or animals, is exploited to move the sink. In direct harvest or the single-hop communication that assumes the sink comes within the direct radio range of all sensors when traveling,

the random mobility leads to incomplete data collection, high delay, and buffer overflow. Because it does not assure that when the sink reaches the vicinity of an arbitrary sensor; consequently, this class is not advised for many real applications, especially those with high sensor count (Kaswan et al. 2018). In multi-hop communication (the sensors that cannot transmit packets to the sink directly, find data routes to do this through some intermediate sensors), random mobility leads to the difficulty of tracking the current position of the sink and increases the routing complexity. Yu et al. (2010) applied the overhearing feature of the wireless networks to track the position of the random mobile sink. They assumed that the mobile sink periodically transmits a beacon message containing its position, and the sensors apply geographic routing. However, the higher rate of sink mobility leads to more energy costs for building, propagating, and maintaining the routes between the sensors and mobile sink (Zhu et al. 2018).

In the case of the predetermined class, the mobile sink is programmed to follow a fixed path in a round-robin manner. This fixed path is predetermined and is not influenced by the sensors' behavior and the environment at run-time. In spite of the inflexibility of this kind of mobility, the sensors have the option to enter the sleep state until the predicted time for data transfer comes (Khan et al. 2013). Chakrabarti et al. (2003) mounted the sink on a city bus, the sensors learn the times that they can connect to the sink and wake up accordingly to transfer their data. In multi-hop communication, the sensors send their data to the nodes located in the vicinity of the sink's future position (Khan et al. 2013). Vajdi et al. (2018) considered a mobile sink that moves in a predefined trajectory in the environment, then select a set of rendezvous points in the periphery of the predefined trajectory. They aimed to minimize the energy consumption of the sensors and reduce the average delay. They proposed a Mixed Integer Linear Programming (MILP) for this problem and presented three distributed heuristic algorithms for planning, identifying rendezvous points, and constructing routes.

In controlled mobility, the sink is moved depending on the network conditions. Two main schemes are applied in most of the existing studies for the controlled class, the trajectory, and itinerary planning. In practice, the application requirements of a WSN specify the appropriate scheme and various approaches have been proposed for each one. The trajectory refers to a non-stop movement in a continuous manner, where the mobile sink can move to any location in the environment, e.g. an Unmanned Aerial Vehicle (UAV) flies above the node's field to harvest the sensed data, or an Autonomous Underwater Vehicle (AUV) is applied in underwater WSNs (Gjanci et al. 2018). In trajectory approaches, the speed variation of the sink is determined by the network load: the sink will move slow if more data should be transmitted by the nearby sensors, fast otherwise.

Zhan et al. (2018) introduced a mixed-integer non-convex optimization problem that jointly optimizes the sensor's wake-up schedule and UAV's trajectory to minimize the maximum energy consumption of all sensors. This model ensures that the required amount of data is collected reliably from each sensor. Tao et al. (2019) proposed an optimization framework that assigns energy for sensors and then finds a trajectory for the mobile sink that covers aggregation nodes. They also developed a heuristic algorithm for the problem.

The itinerary planning determines the sequence of settlement locations where the sink can sojourn at them. Many optimization models have been proposed that provide the itinerary plan of the mobile sink. Almost all optimal approaches gather the network's parameters, then generate an optimization programming, subsequently, provide an optimal itinerary plan. However, the centralized approaches are subject to possible disadvantages like the cost of information gathering, inability to handle dynamic characteristics of WSNs, and need a high degree of complexity (high execution time) (Alsaafin et al. 2018; Yun et al. 2013; Khalily-Dermayn 2021). Moreover, if an error or change occurs in the network (e.g. traffic generation rate changes), a centralized approach would be ineffective because the sink continues on the pre-specified path. In other words, after any change in the network characteristics or environment, a new solution or plan should be provided which imposes a considerable cost to a WSN (Yun et al. 2013). To overcome these drawbacks, researchers in Kaswan et al. (2017, 2018), Khalily-Dermayn and Nadjafi-Arani (2017) assumed high volumes for sojourn time, that is, the sink rarely moves. By this assumption, there exists enough time to gather network parameters and obtain a new solution, while, the delay of reported information increases, especially in single-hop communication.

Site count has a significant influence on the performance of an itinerary plan (Yun and Xia 2010; Roy et al. 2021). In one-hop data collection, a higher count of sojourn sites decreases energy consumption significantly but increases delay and the tour time of the mobile sink. On the contrary, a lower count of sojourn sites may lead to long multi-hop routes and also energy holes. Anwit and Jana (2018) presented a variable-length genetic algorithm to find the count and location of sojourn sites of the sink's tour. To find a trade-off between energy consumption and delay, Roy et al. (2021) studied the optimal count of sojourn sites that guarantee the one-hop data collection from each cluster, then an ant colony optimization algorithm was applied to plan a near-optimal itinerary. In addition, it is proved that the sink travel time plays a major role in the efficiency of mobility and it should be considered as a variable in the models, especially for the slow sink (Keskin and Yiğit 2020).

To simultaneously consider multiple aspects in planning an itinerary, multi-object optimization models have been

introduced by researchers. He et al. (2019) introduced an approach to select potential visiting points and an energy-efficient planning algorithm based on multi-objective particle swarm optimization. This algorithm aims to reduce the delay in data delivery and prolong the network lifetime. To shorten the tour time of the mobile sink, they selected potential visiting points within communication overlapping ranges of sensors, rather than their location. Mehto et al. (2020a) considered data delivery delay, and the traffic rate constraints of sensors during sojourn site selection to design a delay-efficient path for the mobile sink. To meet the delay requirements and minimize energy consumption in industrial WSNs, Ji (2019) proposed a data collection strategy for the mobile sink. They considered the density of sensors, relative residual energy, and the degree of uniformity of distribution, then used the entropy weight method and ant colony optimization algorithm to obtain the optimal access path for the mobile sink. Nitesh and Jana (2019) applied a convex hull for designing a delay-bound path for the mobile sink. This algorithm generates a set of the convex hull as the potential paths and selects one of them as the final path for the mobile sink. Nitesh et al. (2019) considered energy density and some thresholds, then proposed a dynamic, energy-efficient, and delay-bound path for the mobile sink. Moreover, data merit (Gjanci et al. 2018), deadline (Tashtarian et al. 2015), latency (Alsaafin et al. 2018; Alhasanah et al. 2015), density of sensors (Kaswan et al. 2017), and energy consumption (Alsaafin et al. 2018) are of concern in optimizing itinerary plans.

The distributed methods apply local information to obtain an itinerary plan, so they lead to low information gathering cost, ability to handle dynamic characteristics of WSNs, and low execution time (Alsaafin et al. 2018; Yun et al. 2013). Yun et al. (2013) proposed a distributed algorithm that obtains sojourn times, and provides the routing between nodes. This approach limits the maximum tolerable latency. For providing both the sequence of settlement sites and sojourn times, almost all distributed algorithms move greedily the sink towards the best location in the vicinity of its current position. These approaches select a site with the highest volume for an attribute, e.g. a settlement location with more source nodes (Zhu et al. 2016), higher remaining energy (Tan et al. 2009), more traffic load (Akkaya et al. 2005) and so on, subsequently send the sink to that location. If multiple objectives are of concern, the weighted values of the considered attributes are computed and a vicinity location with the highest volume is selected (Tan et al. 2009; Bencan et al. 2020). Wang et al. (2020) proposed a trajectory scheduling method based on coverage rate for multiple mobile sinks in large-scale WSNs. In particular, they used a modified particle swarm optimization to search the sojourn sites with optimal coverage rate, then the genetic algorithm was adopted to schedule the moving trajectory for multiple

mobile sinks. Raj et al. (2020) combined the optimal game theory in selecting the best sojourn sites and ant colony optimization in choosing the best path for the mobile sink to gather data in delay-inevitable WSNs. This distributed algorithm considered the data transfer and management, energy consumption, and delay in data delivery. Alsaafin et al. (2018) proposed a distributed data gathering protocol using a mobile sink for WSN, in which every sensor in the network independently selects itself as a rendezvous point according to a cluster-based method. However, the priority and deadline of the gathered data are not considered by this study. In the best knowledge of the authors, almost all of these distributed methods make greedy decisions in moving the sink, therefore they are not robust, may yield very poor quality solutions, easily fall into the local optima, and may decide that the sink never sojourns at some settlement sites.

To consider multiple criteria in moving the sink, some studies combined itinerary planning with the clustering methods. In these researches, first, the network is divided into some clusters where multiple criteria are of concern, then a path is defined to collect data from cluster heads. Gupta and Saha (2020) focused on a load-balanced clustering scheme that decreases the number of left-out nodes, and then an artificial bee colony algorithm was used for the dynamic moving of the mobile sink. Bencan et al. (2020) presented an evolutionary game-based trajectory algorithm for the mobile sink in WSNs. In their evolutionary game model, the average residual energy of each cluster, and average inter-cluster energy consumption were applied. In particular, the mobile sink moves to a cluster that has the largest residual energy and the shortest distance to other clusters. Preeth et al. (2020) used Adaptive Neuro-Fuzzy Inference System (ANFIS) for optimal cluster head selection where the three input parameters (residual energy, neighbor node sharing, and node behavior history) are of concern. They also employed emperor penguin optimization to find the sojourn sites and traveling paths for the mobile sink. Wang et al. (2018) considered the distance between cluster heads, then an improved ant colony algorithm was applied to find a mobility trajectory for the mobile sink to communicate with cluster heads. Park et al. (2020) proposed an iterative algorithm to minimize the amount of energy consumption. This algorithm performs two phases (clustering and itinerary planning) repeatedly until the amount of energy consumed by the WSN is not reduced. In Sapre and Mini (2021), some relay nodes were deployed using a differential moth flame optimization algorithm, then a traversal strategy for the mobile sink was proposed to collect the sensed data. Anwit et al. (2020) presented a scheme comprising of two algorithms based on the shark smell optimization technique that solves the itinerary planning. First, a non-linear programming and an algorithm were proposed to determine an optimal number of rendezvous points as well as their

locations. Then, a non-linear programming and an algorithm were proposed to find the optimal count of mobile sink and to minimize the overall path length.

Finally, there are some studies on the existing state-of-the-art mobile sink approaches that focus on issues related to sink mobility such as mobility-aware energy-efficient data gathering (Gupta and Gupta 2016), mobility-aware medium access control (Zareei et al. 2018), mobility-aware computational offloading in mobile edge networks (uz Zaman et al. 2021), etc. In addition, Agarwal et al. (2021) presented a survey on the various path selection algorithm used for the mobile sink to collect the information. They also explained some other factors that affect the itinerary planning for a mobile sink. Furthermore, Mehto et al. (2020b) presented the issues of providing a trade-off between energy consumption and data latency.

3 System model and preliminaries

The system model, notations, and the problem statements for moving the mobile sink are presented in this section. In addition, we define some fuzzy definitions applied in the proposed algorithm.

3.1 System model

Let set $\mathbb{V}$ be the set of heterogeneous sensors, scattered in the monitored environment in a random manner. In the network lifetime, the sink travels and sojourns in a set of sites denoted by $\mathbb{L}$. Let $\mathbb{V}^a \subseteq \mathbb{V}$ be the set of sensors around the site $l^a \in \mathbb{L}$ and if the sink arrives at that site, it harvests the stored data from the sensors belongs to $\mathbb{V}^a$. Accordingly, binary parameter $VI_{(q,a)}$ is defined which is 1 if $s_q \in \mathbb{V}^a$; 0 otherwise. In practice, the set $\mathbb{V}$ is partitioned to some subsets in a way that each node belongs to at least one subset, i.e.

$$\bigcup_{l^a \in \mathbb{L}} \mathbb{V}^a = \mathbb{V} \quad (1)$$

Assume that the length of network lifetime is divided into some time-slots numbered from 0 to C where $c \in [0, C]$ is a time-slot of sojourning the sink at an arbitrary site and the last time-slot the sink can receive data is C . The sensors perform continuous sensing, where the sensor s_q senses data $D_{(d,q)}^c$ at the time-slot c . The sensors store the sensed data in their memory until the mobile sink passes by. The attributes of data $D_{(d,q)}^c$ are tabulated in Table 1 and the set of all sensed data that can be delivered to the sink is defined by $\mathbb{D} = \{D_{(d,q)}^c \mid ED_{(d,q)}^c < C\}$.

The sink starts from a fictitious site called l^0 and then sojourns in the sites. The sink sojourns at an arbitrary site

Table 1 The attributes of data and their descriptions

Notation	Definition
$MD_{(d,q)}^c$	The merit of data $D_{(d,q)}^c$
$SD_{(d,q)}^c$	The length of data $D_{(d,q)}^c$
$ED_{(d,q)}^c$	The time-slot that data $D_{(d,q)}^c$ is expired due to deadline
$BD_{(d,q)}^c$	The time-slot that data $D_{(d,q)}^c$ is completely stored in the sensor's memory and it is ready to deliver to the sink

for at least one time-slot, then decides where to go. If the sink currently sojourns in the site $l^a \in \mathbb{L}$, it can go to one of the sites that belongs to:

$$\mathbb{A}^a = \{l^b \mid TT_{(a,b)} \leq TT_{\max}\} \quad (2)$$

where $TT_{\max}$ is the maximum travel distance that the sink is allowed to travel each time it moves and $TT_{(a,b)}$ is the Euclidean distance between the sites l^a and l^b . The set of all possible movements from site l^a is defined by $\mathbb{M}^a = \{(a,b) \mid l^b \in \mathbb{A}^a\}$ and $\mathbb{M} = \bigcup_{l^a \in \mathbb{L}} \mathbb{M}^a$ is the set of pairs that joining sites whose distance of them are less or equal than $TT_{\max}$. At the end of the itinerary plan, the sink travels from any site to another fictitious site called l^{C+1} or the final destination.

The variables of the problem of planing an optimal itinerary for the mobile sink are tabulated in Table 2.

3.2 Objectives

The considered objectives for planning an optimal itinerary and their formulas are as follows:

Data delivery: The volume of delivered data to the sink is an important attribute for planning an itinerary which can be formulated as follows:

$$\text{maximize: } \mathcal{F}_1 = \sum_{D_{(d,q)}^c \in \mathbb{D}} SD_{(d,q)}^c g d_{(d,q)}^c \quad (3)$$

Deadline: Based on what is sensed and/or other properties of data, a deadline is defined for the stored data. The following objective minimizes the volume of expired data.

$$\text{minimize: } \mathcal{F}_2 = \sum_{D_{(d,q)}^c \in \mathbb{D}} SD_{(d,q)}^c e d_{(d,q)}^c \quad (4)$$

Memory: The memory capacity of the sensors is usually limited, so they can not store the collected data for a long time (specifically, if the time interval between visitations is too long). If when data $D_{(d,q)}^c$ is sensing, the memory is full, then it will be lost, so the following objective should be of concern.

Table 2 Variables and their descriptions

Notation	Definition
$ti_{(d,q)}^c$	The time-slot that data $D_{(d,q)}^c$ is delivered to the sink
$gd_{(d,q)}^c$	Binary variable taking the value 1 if data $D_{(d,q)}^c$ is successfully delivered to the sink; 0 otherwise
$ld_{(d,q)}^c$	Binary variable taking the value 1 if a data $D_{(d,q)}^c$ is lost due to memory fullness; 0 otherwise
$ed_{(d,q)}^c$	Binary variable taking the value 1 if a data $D_{(d,q)}^c$ is expired due to its deadline; 0 otherwise
$mo_{(a,b)}^c$	Binary variable taking the value 1 if the sink departs from (leaves) site l^a at time-slot c directed towards l^b ; 0 otherwise
bs_a^c	Binary variable taking the value 1 if the sink is at site l^a at time-slot c ; 0 otherwise
bn_q^c	Binary variable taking the value 1 if the sink is at a site in the vicinity of sensor s_q at time-slot c ; 0 otherwise
um_q^c	The volume of used memory of sensor s_q at time-slot c

$$\text{minimize: } \mathcal{F}_3 = \sum_{D_{(d,q)}^c \in \mathbb{D}} SD_{(d,q)}^c ld_{(d,q)}^c \quad (5)$$

Data Merit: Data merit can be defined through the benefits obtained by taking an action therein, that is, if an urgent event in the environment is reported, the cost of this event and further damages decrease. This merit depends on the captured information and maximizing summation of data merits is one of the considered objectives.

$$\text{maximize: } \mathcal{F}_4 = \sum_{D_{(d,q)}^c \in \mathbb{D}} MD_{(d,q)}^c gd_{(d,q)}^c \quad (6)$$

Latency: The delay tolerant applications can endure the possibly very high latencies imposed by the sink, but the gathered data can not stay in the sensor's memory for a quite long time which imposes very high overall latency. This decreases the count and the type of applications that could apply the mobile sink for data collection. In practice, $ti_{(d,q)}^c - BD_{(d,q)}^c$ defines the latency of data $D_{(d,q)}^c$ and the following objective minimizes the latency of gathered data.

$$\text{minimizes: } \mathcal{F}_5 = \sum_{D_{(d,q)}^c \in \mathbb{D}} \left(ti_{(d,q)}^c - BD_{(d,q)}^c \right) \quad (7)$$

Time to travel: If the sink commutes among far sites, the travel time increases considerably, which affects the network performance. The count of times that sink is commuted between sites l^a and l^b is equal to $\sum_{c \in [0, C]} mo_{(a,b)}^c$ and the following objective minimizes the traveled distance of the sink.

$$\text{minimize: } \mathcal{F}_6 = \sum_{(a,b) \in \mathbb{M}} \left[TT_{(a,b)} \sum_{c \in [0, C]} mo_{(a,b)}^c \right] \quad (8)$$

Fairness: The count of times the sink visits the sensors is another attribute considered in this article. In practice, this objective is applied to balance the count of visitations among the sensors and increases fairness. This goal is defined as the variance minimization among the count of nodes' visitations as follows:

$$\text{minimize: } \mathcal{F}_7 = \text{Var}_{s_q \in \mathbb{V}} \left(\sum_{c \in C} bn_q^c \right) \quad (9)$$

3.3 Constraints

Constraints of optimization problem for obtaining multi-objective-itinerary plan in heterogeneous WSNs are defined as follows:

$$\sum_{l^a \in \mathbb{L}} bs_a^c = 1 \quad \forall c \in [0, C], \quad (10)$$

$$\sum_{l^a \in \mathbb{L}} mo_{(0,a)}^0 = 1, \quad (11)$$

$$\sum_{l^b \in \mathbb{L}} mo_{(b,C+1)}^{C+1} = 1, \quad (12)$$

$$TT_{(a,b)} mo_{(a,b)}^c \leq TT_{\text{Max}} \quad \forall l^a, l^b \in \mathbb{L}, c \in [0, C], \quad (13)$$

$$Ar_a^c = \sum_{\forall (b,a) \in \mathbb{M}^b} mo_{(b,a)}^c \quad \forall l^a \in \mathbb{L}, c \in [0, C], \quad (14)$$

$$Li_a^c = \sum_{\forall (a,b) \in \mathbb{M}^a} mo_{(a,b)}^c \quad \forall l^a \in \mathbb{L}, c \in [0, C], \quad (15)$$

$$mo_{(0,a)}^0 + \sum_{c \in [0, C]} Ar_a^c = mo_{(a,C+1)}^{C+1} + \sum_{c \in [0, C]} Li_a^c \quad \forall l^a \in \mathbb{L}, \quad (16)$$

$$Ar_a^c + Li_a^c \leq bs_a^c \quad \forall l^a \in \mathbb{L}, c \in [0, C], \quad (17)$$

$$Ar_a^c - Li_a^c + bs_a^{c-1} = bs_a^c \quad \forall l^a \in \mathbb{L}, c \in [0, C], \quad (18)$$

$$bn_q^c = bs_a^c * VI_{(q,a)}, \quad \forall s_q \in \mathbb{V}, c \in [0, C], \quad (19)$$

$$um_q^0 = 0, \quad \forall s_q \in \mathbb{V}, \quad (20)$$

$$um_q^{c-1} + SD_{(d,q)}^c = um_q^c + ld_{(d,q)}^c SD_{(d,q)}^c, \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (21)$$

$$um_q^{c+1} = um_q^c - ed_{(d,q)}^c SD_{(d,q)}^c, \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (22)$$

$$um_q^c \leq M_{\text{size}}(1 - bn_q^c), \quad \forall s_q \in \mathbb{V}, c \in [0, C], \quad (23)$$

$$BD_{(d,q)}^c \leq ti_{(d,q)}^c \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (24)$$

$$ti_{(d,q)}^c \leq ED_{(d,q)}^c gd_{(d,q)}^c + BD_{(d,q)}^c [ld_{(d,q)}^c + ed_{(d,q)}^c] \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (25)$$

$$ti_{(d,q)}^c = \min_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} nb n_q^n + ed_{(d,q)}^c BD_{(d,q)}^c \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (26)$$

$$gd_{(d,q)}^c + ld_{(d,q)}^c + ed_{(d,q)}^c = 1 \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (27)$$

$$ed_{(d,q)}^c = \delta \left(\left(1 - ld_{(d,q)}^c \right) \left(1 - \sum_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} bn_q^n \right) \right) \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (28)$$

$$gd_{(d,q)}^c = \delta \left(\left(1 - ld_{(d,q)}^c \right) \sum_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} bn_q^n \right) \quad \forall D_{(d,q)}^c \in \mathbb{D}, \quad (29)$$

$$bn_q^c, bs_a^c, mo_{(a,b)}^c, gd_{(d,q)}^c, ld_{(d,q)}^c, ed_{(d,q)}^c \in \{0, 1\}, um_q^c, ti_{(d,q)}^c \in \{0, 1, 2, \dots\} \quad (30)$$

Constraints (10) enforce that the sink is at exactly one place at any given time-slot c , that is, constraints (10) forbid $bs_a^c = bs_b^c = 1$ for all $l^a, l^b \in \mathbb{L}$, where $l^a \neq l^b$.

Constraints (11) and (12) imply that the sink can start and finish its itinerary plan from/at any arbitrary site. Constraint (11) shows when the itinerary planning is starting (time-slot 0), then the sink travels from the imaginary initial site l^0 to a site $l^a \in \mathbb{L}$. In a similar manner, constraint (12) demonstrates that the sink travels from the last sojourning site l^a to fictitious site l^{C+1} . Constraints (13) restrict the maximum distance that the sink can travel and prevent the latency of long sink traveling.

The auxiliary variables Ar_a^c and Li_a^c are defined by equations (14) and (15), respectively, and they are used in defining constraints (16)–(18). In practice, if an arrival to site l^a from any other site at time-slot c occurs, then $Ar_a^c = 1$ and if the sink departs from site l^a at time-slot c , then $Li_a^c = 1$. Constraints (16) demonstrate that the count of arrivals to the

site l^a ($\sum_{c \in [0, C]} Ar_a^c + mo_{(0,a)}^0$) and the count of departures from the site l^a ($\sum_{c \in [0, C]} Li_a^c + mo_{(0,C+1)}^{C+1}$) are equal.

To demonstrate constraints (17), we consider two cases: (1) if $bs_a^c = 0$, then $Ar_a^c + Li_a^c = 0$, that is, neither a departure from nor an arrival to site l^a during time-slot c occurs; (2) if $bs_a^c = 1$, then $Ar_a^c + Li_a^c \leq 1$, that is, for any site l^a and time-slot c , the sink can select at most one destination for a departure from site l^a (i.e. $0 \leq Ar_a^c \leq 1$) or at most one arrival to site l^a (i.e. $0 \leq Li_a^c \leq 1$), and not both. In addition, the second case ensures that once the sink arrives at site l^a , it cannot leave immediately. Constraints (18) ensure that bs_a^c correctly tracks the sink through time and can be interpreted by considering the following four cases:

- If there are no arrival or departure (i.e. $Ar_a^c = Li_a^c = 0$), then $bs_a^c = bs_a^{c-1}$, so if the sink was at site l^a at time-slot $c - 1$, then it stays at there at time-slot c and otherwise, it is not there in time-slot $c - 1$ and c .
- If there is an arrival (i.e. $Ar_a^c = 1$), then constraints (17) impose that $Li_a^c = 0$ and $bs_a^c = 1$, so $bs_a^{c-1} = 0$, that is, an arrival to site l^a at time-slot c is happened.
- If there is a departure (i.e. $Li_a^c = 1$), then constraints (17) impose that $Ar_a^c = 0$ and $bs_a^{c-1} = 1$, so $bs_a^c = 0$, that is, if the sink is at site l^a at time $c - 1$ and it leaves site l^a at time-slot $c - 1$, then it is not at site l^a at time-slot c .
- The both bs_a^{c-1} and Ar_a^c cannot be 1, because if $Ar_a^c = 1$, then constraints (17) imply that $Li_a^c = 0$, so if $bs_a^{c-1} = 1$ and $Ar_a^c = 1$, then $bs_a^c = 2$, which is impossible for binary variable bs_a^c .

Constraints (19) show that if the sink is at site l^a during the time-slot c (i.e. $bs_a^c = 1$), then it can meet all sensors $s_q \in A^a$ (i.e. $bn_q^c = 1$) and vice versa (if $bs_a^c = 0$, then for all $s_q \in A^a$, $bn_q^c = 0$).

Constraints (20) imply that the sensors' memories are empty at the first time-slot 0. Constraints (21) impose that data $D_{(d,q)}^c$ is either stored in memory (if the memory is not full) or lost due memory fullness (i.e. $ld_{(d,q)}^c = 1$). Constraints (22) imply that if data $D_{(d,q)}^c$ expire, then the sensor s_q deletes that data from its memory. Constraints (23) can be interpreted by two cases: 1) if $bn_q^c = 0$, then no delivery has happened and the used memory of sensors is lower than the maximum capacity of memory M_{size} ; 2) if $bn_q^c = 1$, then data should be delivered and memory is empty.

Constraints (24) show that the time-slot that data $D_{(d,q)}^c$ can be delivered to the sink is after its sensing time-slot. To demonstrate constraints (25) consider two cases: 1) if $gd_{(d,q)}^c = 1$, then $ti_{(d,q)}^c \leq ED_{(d,q)}^c$, that is, data $D_{(d,q)}^c$ is delivered

ered to the sink before its deadline; 2) if either $ld_{(d,q)}^c = 1$ or $ed_{(d,q)}^c = 1$ (constraints (27) impose at most one of them should be 1), then data $D_{(d,q)}^c$ is either lost or expired and the sink can not gather that data, where, constraints (25) imply that $ti_{(d,q)}^c = c$. In practice, the second case demonstrates that if a piece of data is lost or expired, then gathering time is equal to its sensation time-slot, so the lost or expired data do not affect latency. Constraints (26) demonstrate that the time-slot that data $D_{(d,q)}^c$ is delivered to the sink is equal to the first time-slot before its deadline that the sink meets its site. If the sink does not meet the sensor's site before data deadline, then that data is lost and $\min_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} nb_n^n = 0$, here it is assumed that $ti_{(d,q)}^c = c$.

Constraints (27) demonstrate that data $D_{(d,q)}^c$ should be either $gd_{(d,q)}^c = 1$ (i.e. delivered to the sink) or $ld_{(d,q)}^c = 1$ (i.e. lost) or $ed_{(d,q)}^c = 1$ (i.e. expired), while more than one should not be 1. Constraints (28) demonstrate the condition of expiring a data, where,

$$\delta(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases} \quad (31)$$

In particular, constraints (28) is interpreted by the following two cases: (1) if either data $D_{(d,q)}^c$ is lost ($ld_{(d,q)}^c = 1$) or the sink meets the sensor before data deadline ($\sum_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} bn_q^n \geq 1$), then $ed_{(d,q)}^c = 0$; (2) if data $D_{(d,q)}^c$ is not lost ($ld_{(d,q)}^c = 0$) and the sink does not meet the sensor before data deadline ($\sum_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} bn_q^n = 0$), then that data is expired (i.e. $ed_{(d,q)}^c = 1$). Constraints (29) show that (1) if either data $D_{(d,q)}^c$ is lost ($ld_{(d,q)}^c = 1$) or the sink does not meet sensor before data deadline ($\sum_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} bn_q^n = 0$), then $gd_{(d,q)}^c = 0$, that is, this data can not deliver properly; (2) if data $D_{(d,q)}^c$ is not lost ($ld_{(d,q)}^c = 0$) and the sink meets sensor before data deadline ($\sum_{n \in [BD_{(d,q)}^c, ED_{(d,q)}^c]} bn_q^n \geq 1$), then that data is delivered to the sink (i.e. $gd_{(d,q)}^c = 1$). Finally, constraints (31) show the domain sets of the variables.

The optimization problem (3)–(30) is a multi-objective IQCQP and named OP_Sink. Many algorithms can be applied in solving this optimization model, but IQCQPs are computationally NP-hard and cannot be solved in a polynomial-time (Chansombat et al. 2019; Khalily-Dermamy and Nadjafi-Arani 2019). In addition, it needs to gather attributes' volumes into a central node, and then provide the optimal solution. If the network's parameters change, then this process (gathering and solving) should be repeated every time that the sink wants to travel. Therefore, OP_Sink undergoes a noticeable cost for gathering the attributes' volumes and solving the emerged optimization problem. Two options can facilitate these issues: (1) By applying a great

volume of time-slot's period, that is, the sink rarely moves (Kaswan et al. 2017, 2018; Khalily-Dermamy and Nadjafi-Arani 2017). If the sensors store the sensed data until the sink meets their sites, then this great period deteriorates the network performance (lose data, $\mathcal{F}_2$, expired data, $\mathcal{F}_3$, and data latency, $\mathcal{F}_4$, increase, considerably). (2) By generating homogeneous traffic, where, the traffic factors (the size, interval, merit, and deadline of generated data) are fixed. In this situation, the attributes' volumes are predictable over-time and can be computed. Due to the cost of gathering attributes' volumes and solving the optimization problem, it can be concluded that this optimization problem is not beneficial in dynamic and heterogeneous WSNs. However, the ideal itinerary plan determined by solving this optimization problem helps assess the itinerary plan obtained by the proposed algorithm described in the following section.

3.4 Fuzzy definitions

The importance of using a fuzzy TOPSIS is to assign the relative importance of attributes using fuzzy numbers instead of precise numbers and have discussed in (Salih et al. 2019). Some useful definitions of fuzzy concepts relevant to understanding the proposed algorithm are described in the following [for more detail see Salih et al. (2019)].

Definition 1 A fuzzy set is a pair (X, μ) where X is a set and $\mu : X \rightarrow [0, 1]$ is a membership function. The reference set X is named universe of discourse and for each $x \in X$, the value $\mu(x)$ is named the grade of membership of $x \in X$. The function $\mu(x)$ maps each element $x \in X$ to a real number in the interval $[0, 1]$.

Definition 2 A triangular fuzzy number is represented by a triplet $\tilde{x} = (x', x'', x''')$ where x', x'' and x''' are the real numbers and $x' \leq x'' \leq x'''$ holds true. The membership function $\mu(x)$ of triangular fuzzy number $\tilde{x}$ is expressed as:

$$\mu(x) = \begin{cases} \frac{x-x'}{x''-x'} & \text{if } x' \leq x \leq x'' \\ \frac{x'''-x}{x'''-x''} & \text{if } x'' \leq x \leq x''' \\ 0 & \text{Otherwise} \end{cases} \quad (32)$$

The value of $\tilde{x}$ at x'' yields the maximal grade of $\mu(x)$, i.e., $\mu(x'') = 1$. While, the value of x at x' and x''' yields the minimal grade of $\mu(x)$, (i.e., $\mu(x') = 0$ and $\mu(x''') = 0$), indicating that the constants x' and x''' are the lower and upper bounds of the available area.

Definition 3 Let both the $\tilde{x} = (x', x'', x''')$ and $\tilde{y} = (y', y'', y''')$ define the two triangular fuzzy numbers, then the distance between them is expressed as:

$$F_{dis}(\tilde{x}, \tilde{y}) = \sqrt{\frac{1}{3}[(x' - y')^2 + (x'' - y'')^2 + (x''' - y''')^2]} \quad (33)$$

Definition 4 Let both the $\tilde{x} = (x', x'', x''')$ and $\tilde{y} = (y', y'', y''')$ define two triangular fuzzy numbers, then the product between them is expressed as:

$$\tilde{x} \odot \tilde{y} = (x' \cdot y', x'' \cdot y'', x''' \cdot y''') \quad (34)$$

Definition 5 The function Λ converts a set of real numbers, A , to a triangular fuzzy number and returns the three numbers x', x'' and x''' according to

$$\Lambda(A) = (x', x'', x''') = (\min(A), \text{avg}(A), \max(A)) \quad (35)$$

where, $\min(A)$, $\text{avg}(A)$, and $\max(A)$ return the minimum, average, and maximum elements of the set A , respectively.

4 Proposed algorithm

This paper applies the fuzzy TOPSIS to plan an itinerary for the sink, where, some attributes from the sensors and sites are of concern. In brief, it compares the set of possible future sites by identifying their weights and normalizing scores for each attribute. The closest site to the Fuzzy Positive Ideal Solution (FPIS) and farthest from the Fuzzy Negative Ideal Solution (FNIS) is selected as the most appropriate alternatives. A trade-off among the attributes where a poor outcome can be negated by a good result in another one is made through this algorithm.

The pseudo-code of this proposed algorithm is expressed in Algorithm 1, named MA_Sink. This algorithm assumes that the sink currently sojourns in the site l^a and wants to select the next site from $|A^a|$ adjacent sites by considering the attributes $C_1, \dots, C_8 \in J$. In the following, the steps of MA_Sink are demonstrated.

Algorithm 1 The pseudo code of MA_Sink.

```

1: procedure ITINERARY_PLANNING()
2:    $D\tilde{M}_1 \leftarrow \text{INITIALIZE}()$  ;
3:    $D\tilde{M}_2 \leftarrow \text{NORMALIZE}(D\tilde{M}_1)$  ;
4:    $D\tilde{M}_3 \leftarrow \text{WEIGHTING}(D\tilde{M}_2, W)$  ;
5:    $[F^+, F^-] \leftarrow \text{FPIS\_FNIS}(D\tilde{M}_3)$  ;
6:    $[d^+, d^-] \leftarrow \text{DIST}(D\tilde{M}_3, F^+, F^-)$  ;
7:    $CL \leftarrow \text{CLOSENESS}(d^+, d^-)$  ;
8: end procedure

```

4.1 Attributes

In selecting the most appropriate site from the adjacent sites, 8 attributes $C_1, \dots, C_8 \in J$ are applied in this article, however, this proposed algorithm can consider more attributes for future applications. A brief description and the impacts of the attributes are tabulated in Table 3. These attributes comprise two subsets, beneficiary and cost attributes. First, let $J^+ = \{C_1, C_3, C_4, C_5, C_8\}$ be the set of beneficiary attributes with positive impact where a site with higher values for the volume of delivered data (C_1), used memory (C_3), data merit (C_4), latency (C_5), and the count of active sensors (C_8) is more appropriate to be the next sojourn site. On the other hand, the lower values of non-beneficial or cost attributes which constitute J^- are desirable. More clearly, a site with lower values for the deadline (C_2), travel time (C_5),

and count of visits (C_6) is more preferable. It is obvious that the union of positive and negative attributes is equal to J (i.e. $J = J^+ \cup J^-$).

Table 3 The attributes index, description and impact

Index	Description	Impact
C_1	Data delivery	+
C_2	Data deadline	−
C_3	Used memory	+
C_4	Merit of data	+
C_5	Latency	+
C_6	Time to travel	−
C_7	Count of visits	−
C_8	Count of active sensors	+

4.2 Initialization

When the sink wants to move, it broadcasts a request message for obtaining the volumes of the considered attributes of each site in its vicinity. It is assumed that there exists a sentinel node for each site that will be in charge of reporting the attributes of its site. The task of sentinel nodes is to collect and report the attributes' volumes from their vicinity when requested by the sink, that is, the sink interrogates the sentinel nodes of adjacent sites to collect their attributes. The sentinel nodes can be selected by the sink or elected by the sensors surrounding their sites.

In particular, the first step of MA_Sink consists of the following four phases.

Phase 1: The sink broadcasts a (beacon) packet. The sensors in the transmission domain of the sink can receive this packet.

Phase 2: If a sentinel node receives the beacon, then collects 8 sets of real numbers corresponding to the attributes from its neighbors. In practice, this phase is accomplished by querying the neighbor sensors about their characteristics by the sentinel nodes. Then, the function Λ is applied to convert the collected sets into the triangular fuzzy numbers.

Phase 3: The sentinel nodes communicate back to the sink the fuzzy converted of the collected sets. As to the 8 attributes, each sentinel node transmits 8 triangular fuzzy numbers to the sink. This mechanism allows the sink to identify the accessible sites, because no packet is received from the sensors from the non-accessible sites.

Phase 4: The sink collects the attributes of all accessible sites and generates the fuzzy decision matrix $\tilde{DM}_1$ as follows:

$$\tilde{DM}_1 = \begin{matrix} & C_1 & C_1 & \dots & C_8 \\ \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_{|\mathbb{A}^a|} \end{matrix} & \begin{pmatrix} \tilde{x}_{1,1} & \tilde{x}_{1,2} & \dots & \tilde{x}_{1,8} \\ \tilde{x}_{2,1} & \tilde{x}_{2,2} & \dots & \tilde{x}_{2,8} \\ \vdots & \vdots & \tilde{x}_{i,j} & \vdots \\ \tilde{x}_{|\mathbb{A}^a|,1} & \tilde{x}_{|\mathbb{A}^a|,2} & \dots & \tilde{x}_{|\mathbb{A}^a|,8} \end{pmatrix} \end{matrix} \quad (36)$$

where, $\tilde{x}_{ij} = (x'_{ij}, x''_{ij}, x'''_{ij})$ is the triangular fuzzy number that specifies the appropriateness of site A_i corresponding to the attribute C_j .

4.3 Normalization

Normalization is usually required because the attributes are often of incongruous dimensions. In this article, the linear scale transformation is applied to convert different attributes scales into a comparable scale. The normalized matrix is named $\tilde{DM}_2 = [\tilde{r}_{ij}]_{|\mathbb{A}^a| \times 8}$, where, $\tilde{r}_{ij}$ is a triangular fuzzy

number. For the benefit attributes (the C_1, C_3, C_4, C_5 , and C_8 that belong to J^+) $\tilde{r}_{ij}$ is defined as follows:

$$\tilde{r}_{ij} = \left(\frac{x'_{ij}}{c_j^+}, \frac{x''_{ij}}{c_j^+}, \frac{x'''_{ij}}{c_j^+} \right) \quad (37)$$

where, $c_j^+ = \max_{i=1 \dots |\mathbb{A}^a|} x'''_{ij}$. For the cost attributes (the C_2, C_6 , and C_7 that belong to J^-) $a_j^- = \min_{i=1 \dots |\mathbb{A}^a|} x'_{ij}$ is defined as follows:

$$\tilde{r}_{ij} = \left(\frac{a_j^-}{x'''_{ij}}, \frac{a_j^-}{x''_{ij}}, \frac{a_j^-}{x'_{ij}} \right) \quad (38)$$

Note that the mentioned normalization method maintains the normalized triangular fuzzy numbers within $[0, 1]$ range.

4.4 Weighting the normalized decision matrix

The weighted normalized decision matrix, $\tilde{DM}_3 = [\tilde{v}_{ij}]_{|\mathbb{A}^a| \times 8}$, is calculated by multiplying $\tilde{DM}_2$ with the weight $\tilde{W} = [\tilde{w}_1, \dots, \tilde{w}_8]$, as follows:

$$\tilde{v}_{ij} = \tilde{r}_{ij} \odot \tilde{w}_j = (w'_j \cdot r'_{ij}, w''_j \cdot r''_{ij}, w'''_j \cdot r'''_{ij}). \quad (39)$$

where, $\tilde{w}_j$ is the attribute C_j 's preference.

4.5 Determining FPIS and FNIS

In the fuzzy TOPSIS, the most preferable alternative, FPIS, is consists of the best performance values for each alternative, while, the least preferable alternative, FNIS, consists of the worst performance values. This step obtains $F^+ = (\tilde{f}_1^+, \tilde{f}_2^+, \dots, \tilde{f}_8^+)$ and $F^- = (\tilde{f}_1^-, \tilde{f}_2^-, \dots, \tilde{f}_8^-)$ which correspond to FPIS and FNIS, respectively. In determining $\tilde{f}_j^+$ and $\tilde{f}_j^-$, the cost and benefit attributes are different. In the benefit attributes, $\tilde{f}_j^+ = (c, c, c)$ and $\tilde{f}_j^- = (a, a, a)$, while, for the cost attributes that belong to the J^- , $\tilde{f}_j^+ = (a, a, a)$ and $\tilde{f}_j^- = (c, c, c)$ are of concern, where, $a = \min_i \{v'_{ij}\}$ and $c = \max_i \{v'''_{ij}\}$, respectively.

4.6 Calculating the distance to FPIS and FNIS

In this step, the separation of the existing alternatives from FPIS and FNIS is made. It highlights the similarity between alternatives with the best and worst solutions. This distance is calculated by the Euclidean distance.

Let d_i^+ and d_i^- define the distance of alternative A_i from FPIS and FNIS, respectively, expressed as:

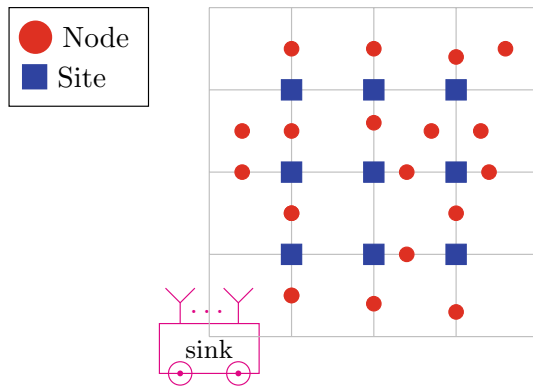


Fig. 1 An instance of the deployment area, where, $N_v = 20$ and $N_s = 3$

$$d_i^+ = \sum_{j=1}^8 F_{dis}(\tilde{v}_{i,j}, \tilde{f}_j^+) \quad (40)$$

$$d_i^- = \sum_{j=1}^8 F_{dis}(\tilde{v}_{i,j}, \tilde{f}_j^-) \quad (41)$$

4.7 Rank the alternatives

The closeness coefficients are calculated to determine the relative closeness to FPIS and FNIS in a simultaneous manner expressed as:

$$CL_i = \frac{d_i^-}{d_i^- + d_i^+} \quad i = 1, \dots, |\mathbb{A}^a| \quad (42)$$

Because $d_i^+ \geq 0$ and $d_i^- \geq 0$, $CL_i \in [0, 1]$. Alternative with the highest CL_i is the best alternative because it is the closest to FPIS and farthest from FNIS. In particular, $CL_i = 1$ if and only if the alternative A_i is in its best condition; and $CL_i = 0$ if and only if it is in its worst condition.

5 Performance evaluation

In this paper, simulations are made to analyze the effectiveness of the proposed methods in WSNs. We assume that N_v sensors are dispersed in a random manner in the environment, and there exist $N_s \times N_s$ sites (as a regular grid). The deployment area where $N_v = 20$ and $N_s = 3$ are shown in Fig. 1. To eliminate the difference caused by the random parameters, simulations are repeated 200 times and the presented results are the average of outcomes of these repetitions. Assume that the closest distance between the sites

Table 4 The lower, average and upper bounds of traffic factors

Factor	Average	Range
Data size	$\overline{SD} = 1 \text{ kB}$	$\left[\frac{\overline{SD}(100-\chi)}{100}, \frac{\overline{SD}(100+\chi)}{100} \right]$
Data inter-arrival	$\overline{ID} = 1 \text{ s}$	$\left[\frac{\overline{ID}(100-\chi)}{100}, \frac{\overline{ID}(100+\chi)}{100} \right]$
Merit	$\overline{MD} = 100$	$\left[\frac{\overline{MD}(100-\chi)}{100}, \frac{\overline{MD}(100+\chi)}{100} \right]$
Deadline	$\overline{DD} = 10 \text{ min}$	$\left[\frac{\overline{DD}(100-\chi)}{100}, \frac{\overline{DD}(100+\chi)}{100} \right]$

is 100 m and the transmission range of sensors is 141 m. Changing N_s is similar to that of varying the environment size with slight differences. The sensors sense the environment and store the gathered data in their memory until the mobile sink meets their sites. The simulations are ran for 100 time-slots ($C = 100$) and the time-slot's period is 60 s. The memory size is 512 kB and the sensors lose the new sensed data in the case of memory fullness. Note that the merit of the stored data does not decay, but if the deadline of a stored data reaches, then its merit is reduced to zero, indicating that there is no interest in data delivery.

The heterogeneity of a WSN depends on its application requirements, which dictate the generated traffic of sensors. In this article, a heterogeneous coefficient named χ is applied to control the lower and upper bounds of the traffic factors. It supports heterogeneous sensing/computing requirements of traffic generation. In practice, the data length, time interval, merit, and deadline of the sensed data are specified by a random distribution over their range in which are tabulated in Table 4. For instance, the size of a generated data is distributed within $\frac{\overline{SD}(100-\chi)}{100}$ to $\frac{\overline{SD}(100+\chi)}{100}$ in a random manner.

The following four methods are compared in this section.

- The sink travels along with the settlement sites with the optimum plan derived by a multi-objective optimization (3)–(30), named the OP_Sink.
- The sink adopts MA_Sink and travels according to its scenario.
- The sink moves towards a site with higher weighted attributes as specified by GR_Sink in a greedy manner. This method is distributive and considers the described attributes.
- The method is termed PM_Sink (for Predetermined Mobility), where the sink travels in cycles around the sites in a fixed and predetermined path.

The results for OP_Sink have been obtained by solving the IQCQP defined in Sect. 3 using Pyomo (Hart et al. 2017) and the CPLEX (CPLEX 2009). MA_Sink (Sect. 4) is implemented in a home-grown software framework by Python 3.9.

All simulations are run on Windows 10 personal computer with an Intel Core i5 CPU 2.53 GHz and 8 GB RAM, then, the obtained network models are implemented in the INET Framework of OMNeT++ (Varga and Hornig 2008).

5.1 Time complexity

The assessed methods have different execution times or computational complexity in obtaining their solutions, which, affects their application in the practice.

The complexity of PR_Sink is lower than other methods because it is not influenced by the network's attributes. In practice, a fixed path is specified, where, the sink travels to all sites in a round-robin manner, consequently, it consumes no energy for collecting the attributes and executes no computation to specify the movement plan.

To decide which one is more appropriate to be the next site, both the GR_Sink and MA_Sink collect the attributes' volumes from the adjacent sites. GR_Sink computes the summation of products of the attributes' volumes with the corresponding weights. If the sink sojourns in the site l^a , then GR_Sink can be executed in $O(|\mathbb{A}^a|)$. The computational complexity of MA_Sink is similar to selecting an alternative by fuzzy TOPSIS where 8 attributes are of concern. If the sink currently sojourns in site l^a , then the computational cost of MA_Sink for choosing the most appropriate adjacent site is $O(|\mathbb{A}^a|)$. This complexity value is obtained from the calculation of normalization and weighting, as well as the complexity in obtaining FPIS and FNIS. Consequently, the computational complexity of both the MA_Sink and GR_Sink belongs to $O(N_s^2)$, because $|\mathbb{A}^a| \leq N_s^2$.

OP_Sink gathers all sites' attributes, then draws the optimal itinerary plan. If the network's parameters change, then this process should be repeated which wastes noticeable energy of sensors. Furthermore, there is not any algorithm

that can solve OP_Sink in polynomial time. In dynamic and heterogeneous WSNs where large number of gathering and solving processes are needed, the OP_Sink burdens noticeable costs to the network in terms of time and energy consumption, so it is impractical and almost impossible in large-scale WSNs.

In the following, the first scenario considers the sensors transmit a homogeneous traffic pattern; the second scenario involves heterogeneous traffic generation where the traffic's factors vary with time in a random manner.

5.2 Homogeneous traffic

If the sensors sense certain data from the environment with a fixed interval, then there is no heterogeneity, $\chi = 0$, and traffic generation is homogeneous. The smart meter devices are the samples of such applications where, they gather consumption, diagnostic, and status data from water or energy (gas or electric) meters in an automated manner.

The effect of varying sensor and site counts on the total delivered data is drawn in Fig. 2. In this simulation, it is assumed that $TT_{\max} = 100$ m and there are two cases for the site count: $N_s = 4$, and $N_s = 8$. Due to growing the generating data, an increase in the sensor count increases the total delivered data. By comparing Fig. 2a, b, it is revealed that an increase in the site count would decrease the total delivered data. Consequently, the objective $\mathcal{F}_1$ has a direct relation with the sensor count, and the opposite holds true with the site count. Indeed, the results of this simulation reveal that OP_Sink and PR_Sink obtain the best and worst outcomes, respectively. In addition, if N_s and N_v increase then the total delivered data of MA_Sink, GR_Sink, and PR_Sink decreases when compared with OP_Sink. In other words, the efficiency of the proposed multi-attribute algorithm in delivering data decreases in the sparser networks

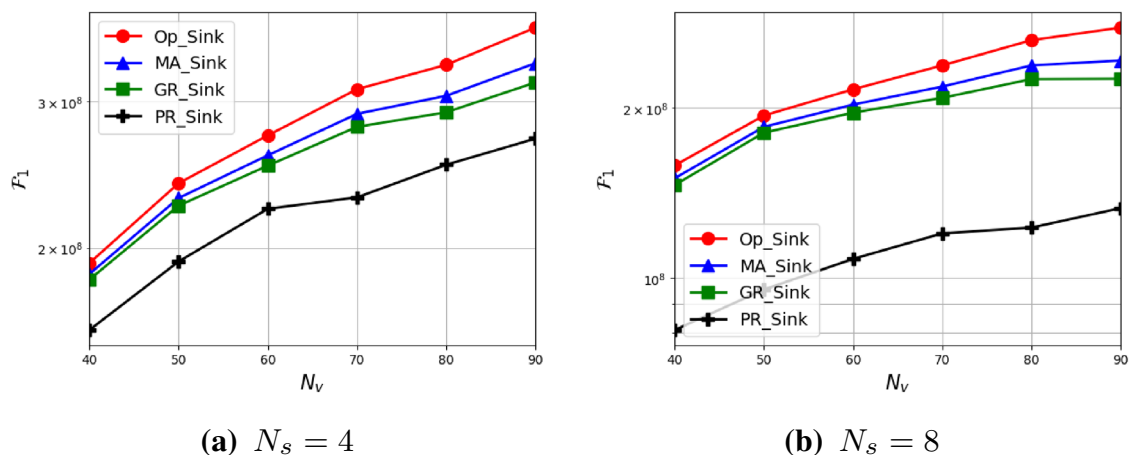


Fig. 2 The effect of changing N_v and N_s on the objective $\mathcal{F}_1$, where $TT_{\max} = 100$ m

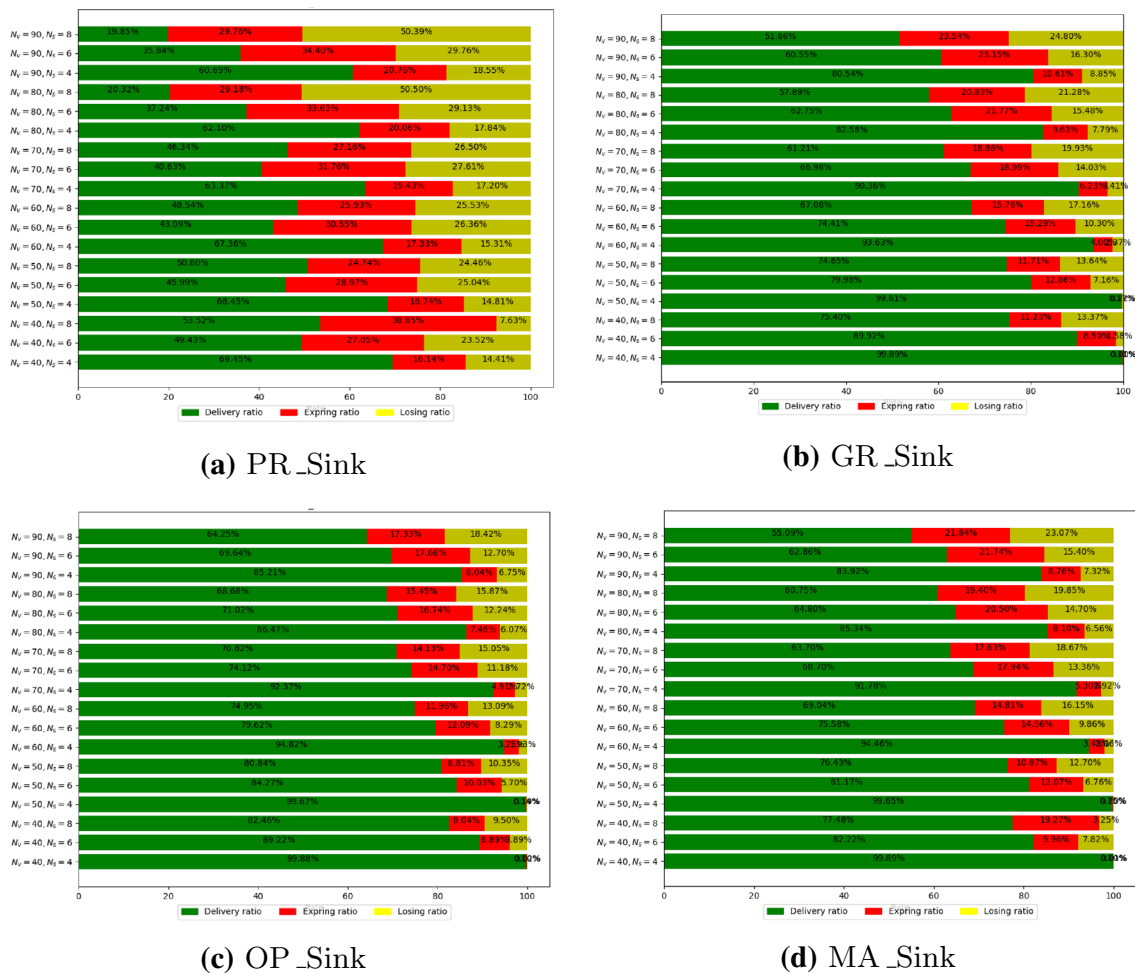


Fig. 3 The effect of varying N_v and N_s on the total delivery, expiring and losing ratio, where $TT_{\max} = 100$ m

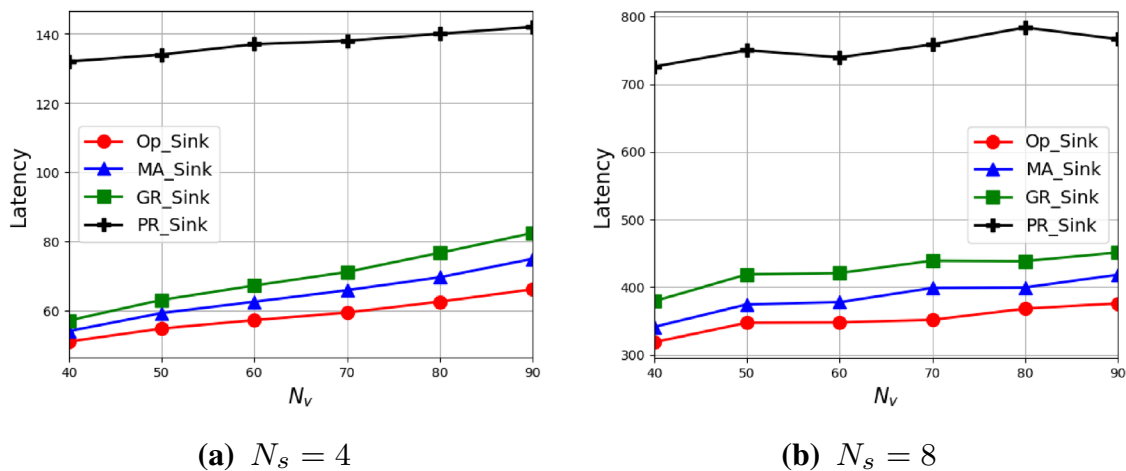


Fig. 4 The average of latencies at different sensor and site counts, where $TT_{\max} = 100$ m

and vast environment. Moreover, MA_Sink outperforms GR_Sink, although this experiment shows their results are

coherent with each other. Not providing any kind of compensation between the adjacent sites is the major drawback of

GR_Sink. Due to considering equal merit for all data (homogeneous traffic generation), the trends of the figures for total data merit are similar.

The effect of varying the sensor and site counts on the delivery, expiring, and losing ratio with respect to the total generated data are reflected in Fig. 3. In this simulation, it is assumed that $TT_{\max} = 100$ m, the sensor count is varied from 40 to 90 (with step 10) and there are three cases for the site count: $N_s = 4$, $N_s = 6$, and $N_s = 8$. The results reveal that OP_Sink provides the best outcomes (higher delivery ratio as well as lower expiring and losing ratio), while PR_Sink provides the worst. In addition, an increase in the site count leads to lower efficiency, because the probabilities of expiring and losing data increase. Furthermore, the gaps among OP_Sink and other methods widen when N_v and N_s increases, indicating that the performance of MA_Sink decreases when site count increases.

The average of latencies when N_v increases from 40 to 90 is drawn in Fig. 4. The simulation parameters are kept at the previously specified volumes and there are two cases for the site count: $N_s = 4$, and $N_s = 8$. The results reveal that an increase in the sensor count leads to lower latency, while an increase in the site count leads to higher latency. Same as the previous simulations, the OP_Sink and PR_Sink obtain the best and worst outcomes, respectively.

The effect of varying sensor and site counts on the objectives $\mathcal{F}_6$ (the sink's traveling distance) and $\mathcal{F}_7$ (fairness) are reflected in Figs. 5 and 6, respectively. In this simulation, it is assumed that $TT_{\max} = 100$ m and there are two cases for the site count: $N_s = 4$, and $N_s = 8$. In all time-slots, PR_Sink moves the sink, so varying the sensor and site counts has no effect on the sink's movement. It is possible that OP_Sink, MA_Sink, and GR_Sink decide the sink stay at a sojourn site for more than one time-slot, so their traveling distance

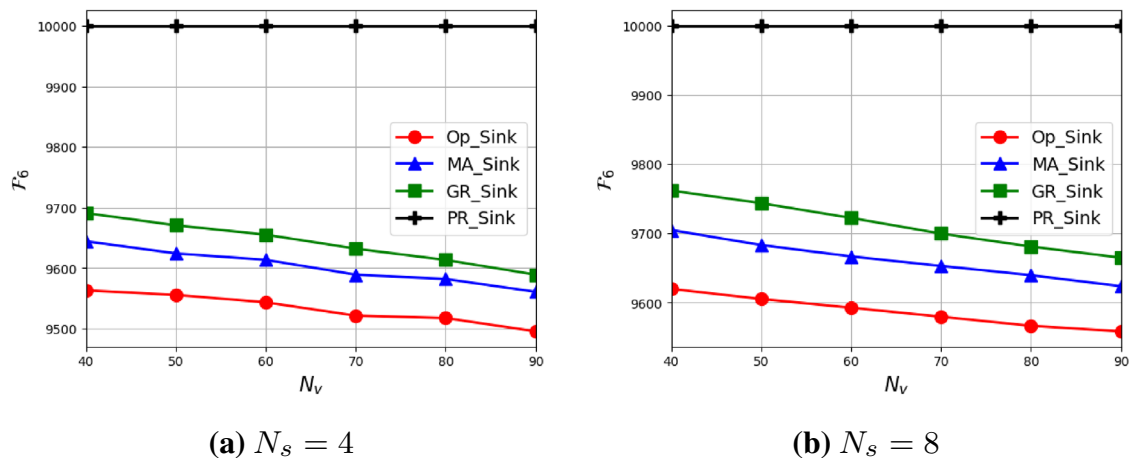


Fig. 5 The effect of changing N_v and N_s on the sink's movement, where $TT_{\max} = 100$ m

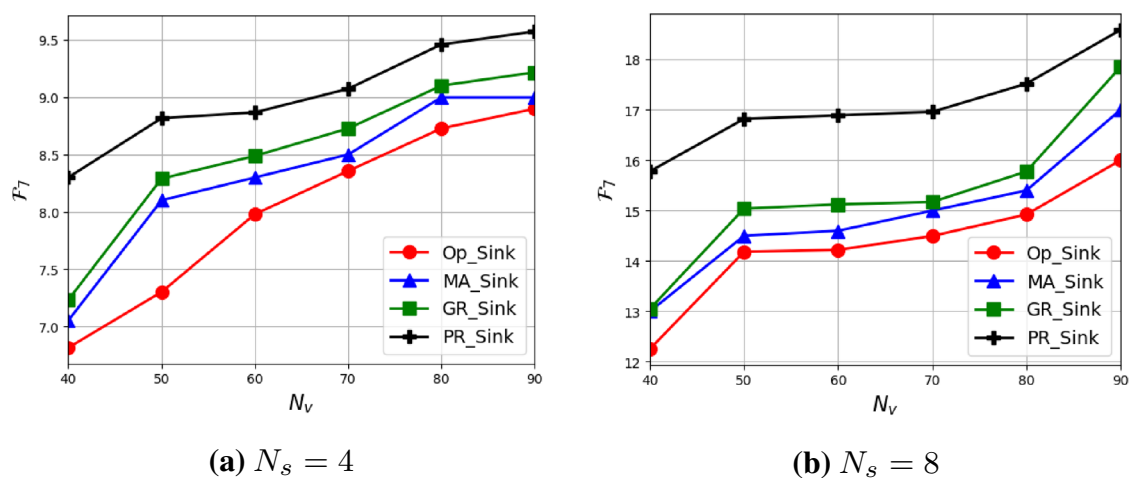


Fig. 6 The effect of changing N_v and N_s on the objective $\mathcal{F}_7$, where $TT_{\max} = 100$ m

is lower than PR_Sink. However, considering low traveling distance, $TT_{\max}$, leads to small differences among methods. Figure 6 shows that the best and worst volumes of $\mathcal{F}_7$ are associated with OP_Sink and PR_Sink, respectively. In particular, the sensors' density is not of concern in PR_Sink, so it yields lower fairness among the sensors. In addition, an increase in the sensor and site counts leads to increasing variance among the visitations' counts and lower fairness.

The simulations results show that OP_Sink outperforms MA_Sink in all objectives $\mathcal{F}_1$ – $\mathcal{F}_7$. The reason for this observation is that MA_Sink gathers attributes' volumes from the sink's neighbor and decides based on, while, the OP_Sink applies global information to plan the itinerary. To assess the efficiency of MA_Sink correspond to OP_Sink, the difference rate $\mathcal{D}_1$ is defined as follows.

$$\mathcal{D}_1 = 100 \times \frac{\mathcal{F}_1(\text{OP_Sink}) - \mathcal{F}_1(\text{MA_Sink})}{\mathcal{F}_1(\text{OP_Sink})}, \quad (43)$$

where $\mathcal{F}_1(\text{OP_Sink})$ and $\mathcal{F}_1(\text{MA_Sink})$ denote the volume of delivered data by OP_Sink and MA_Sink, respectively. Then, the difference rates $\mathcal{D}_3$ – $\mathcal{D}_5$ are defined in a similar manner. For the cost attributes ($\mathcal{D}_2$, $\mathcal{D}_6$ and $\mathcal{D}_7$), we use the following formula.

$$\mathcal{D}_2 = 100 \times \frac{\mathcal{F}_2(\text{MA_Sink}) - \mathcal{F}_2(\text{OP_Sink})}{\mathcal{F}_2(\text{OP_Sink})}, \quad (44)$$

Table 5 The difference rates between OP_Sink and MA_Sink in different N_v and N_s , where $TT_{\max} = 100$ m

		N_ν				
		N_s	50 (%)	60 (%)	70 (%)	80 (%)
$\mathcal{D}_1$	4	0.02	0.38	0.64	1.31	1.51
	6	3.68	5.07	7.31	8.76	9.74
	8	5.46	7.89	10.05	11.55	14.26
$\mathcal{D}_2$	4	6.87	7.11	7.89	8.57	8.98
	6	20.31	20.41	22.02	22.44	23.08
	8	23.38	23.83	24.78	25.54	26.01
$\mathcal{D}_3$	4	6.48	6.84	7.23	8.07	8.45
	6	18.57	18.95	19.54	20.12	21.28
	8	22.67	23.37	24.05	25.07	25.22
$\mathcal{D}_4$	4	0.02	0.38	0.64	1.31	1.51
	6	8.20	5.07	7.31	8.76	9.74
	8	5.46	7.89	10.05	11.55	14.26
$\mathcal{D}_5$	4	2.12	2.54	3.33	3.58	3.98
	6	2.54	2.76	3.71	4.84	5.09
	8	3.27	3.61	4.51	6.05	6.98
$\mathcal{D}_6$	4	0.67	0.68	0.71	0.73	0.77
	6	0.70	0.85	0.88	0.97	1.05
	8	0.81	0.87	0.96	1.08	1.25
$\mathcal{D}_7$	2	1.51	2.49	2.55	2.71	2.85
	3	2.28	2.39	3.05	3.28	3.49
	4	2.98	2.98	3.43	3.95	4.18

Table 6 The differences rates between OP_Sink and MA_Sink in various $TT_{\max}$, where $N_v = 90$ and $N_s = 8$

	$TT_{\max}$			
	100 (%)	200 (%)	300 (%)	400 (%)
$\mathcal{D}_1$	14.26	13.00	11.48	10.03
$\mathcal{D}_2$	26.01	23.76	20.90	18.40
$\mathcal{D}_3$	25.22	23.06	20.58	18.09
$\mathcal{D}_4$	14.26	13.00	11.48	10.03
$\mathcal{D}_5$	6.98	6.33	5.60	4.83
$\mathcal{D}_6$	1.25	1.98	3.21	5.28
$\mathcal{D}_7$	4.18	4.18	4.18	4.18

Clearly, higher difference rates show that OP_Sink acts better than MA_Sink. Table 5 shows the difference rates in different N_v and N_s , where $TT_{\max} = 100$ m. If $N_v = 40$ and $N_s = 4$, then, minimum difference rate $\mathcal{D}_1$ is observed and the maximum $\mathcal{D}_1$ is occurred at a network with $N_v = 90$ and $N_s = 8$. The reason for this observation is that an increase in site count leads to decreasing the proportional the neighbors' sites (where MA_Sink can get the volumes of their attributes) to the total site ratio, while OP_Sink considers global information. The highest gap is occurred in expiring, $\mathcal{D}_2$, and losing data, $\mathcal{D}_3$. In general, the probability of losing and expiring data increases in sparser and vaster networks. As mentioned considering equal merits for all data leads to

having similar $\mathcal{D}_1$ and $\mathcal{D}_4$ in all cases. Furthermore, considering $TT_{\max} = 100$ m leads to lower gaps in traveling distances and consequently lower difference rate $\mathcal{D}_6$.

The difference rates between OP_Sink and MA_Sink in different $TT_{\max}$, where $N_v = 90$ and $N_s = 8$ are tabulated in Table 6. Indeed, an increase in $TT_{\max}$ from 100 m to 400 m would reduce the differences rates $\mathcal{D}_1$ – $\mathcal{D}_5$ and $\mathcal{D}_7$. This is because that an increase in $TT_{\max}$ increases the count of adjacent sites to $N_s \times N_s$ ratio, thus, allowing the MA_Sink to choose the better decision and a decrease in the differences rates. But an increase in $TT_{\max}$ increases $\mathcal{D}_6$, because the sink can travel to further sites and the proposed algorithm may fail to select the optimal path. However, increasing $TT_{\max}$ leads to more sentinel nodes and higher energy consumption which are described in the next subsection.

5.3 Heterogeneous traffic

The OP_Sink is aware of the attributes' volumes of whole sites and sends the sink to the global optimum site in advance. This subsection considers dynamic and heterogeneous WSNs, where the factors of generated traffic (data size, interval, merit, and deadline) are not the same and do not remain constant over time (they change occasionally). Indeed, applying dynamic and heterogeneous traffic pattern affects the attributes' volumes in the process of time and predicting them are not simple.

If the sensors generate dynamic and heterogeneous traffic, then OP_Sink can not gather the attributes' volumes from all sites, because it imposes an additional energy cost on the networks. The details of the total energy consumption for collecting information from the sites in the range R are tabulated in Table 7. The sequence of interpreting this table is as follows:

- The consumed energy for transmission by a sentinel node is expressed through: $E^{tx} = \eta_1 R^\alpha + \eta_2$ where, α is the path loss exponent with $2 \leq \alpha \leq 6$ depending on the environment, η_1 and η_2 are the parameters subject to the characteristic of transceiver and the channel. In the existing wireless transceivers, it can be assumed that $\alpha = 2$,

$\eta_1 = 45$ nJ bit⁻¹ and $\eta_2 = 10$ pJ bit⁻¹ m² (Khalily-Dermany and Nadjafi-Arani 2019).

- If R increases, then the count of adjacent sites increases, that is, more sentinel nodes are responsible to report to the sink. An increase in R from 100 to 400 leads to an increase in the maximum adjacent sites' count, $\max_{\mu \in \mathbb{L}} |\mathbb{A}|^\alpha$, from 5 to 40.

Consequently, we can compute the minimum and maximum energy consumption for collecting the attributes' volumes by the sentinel nodes. Therefore, an increase in R from 100 to 400 m leads to about 90 and 156 fold higher energy consumption. In addition, if $\alpha = 3$ (in the harsh environment), then an increase in R from 100 to 400 m leads to about 362 and 627 fold higher energy consumption. Moreover, if we consider the reception energy, then it is becoming more evident that increasing R has a significant burden on the network and energy consumption.

In dynamic networks, the attributes' volumes change occasionally in the process of time, so OP_Sink should gather all attributes' volumes in all time-slots from the sentinel nodes. This burdens noticeable energy to the sensors, so this article uses the averages of traffic factors for computing the OP_Sink.

The effect of varying sensor and site counts on the total delivered data in heterogeneous traffic is drawn in Figs. 7 and 8 where $\chi = 10$ and $\chi = 20$, respectively. In this simulation, it is assumed that $TT_{\max} = 100$ m and there are three cases for the site count: $N_s = 4$, $N_s = 6$, and $N_s = 8$. Same as homogeneous traffic, an increase in the sensor count increases the total delivered data because more traffic is generated. In addition, an increase in the site count decreases the total delivered data, because more data may be lost or expired. But the results of this simulation reveal that MA_Sink may outperform OP_Sink and deliver more data to the sink. More clearly, if $N_v = 90$, $N_s = 4$ and $\chi = 10$, then MA_Sink overtakes OP_Sink. The reason for this observation is that OP_Sink provides the sink's movement based on preliminary attributes, while the instantaneous information is of concern by MA_Sink. In addition, an increase in sensor count leads to more efficiency for MA_Sink in comparison with OP_Sink, because more heterogeneous and dynamical data is generated, and consequently using local information by MA_Sink leads to its higher efficiency. In practice, an increase in the heterogeneity coefficient (Fig. 8) leads to a more decrease in OP_Sink's performance, so MA_Sink overtakes OP_Sink in lower sensor count than the one shown in Fig. 7. Because higher variety among the attributes in the sites is achieved if an increase in heterogeneity coefficient of the traffic generation occurs.

Table 7 The effect of changing R

	R			
	100	200	300	400
E^{tx}	0.45 mJ	1.80 mJ	4.05 mJ	7.20 mJ
$\min_{\mu \in \mathbb{L}} \mathbb{A} ^\alpha$	3	6	11	17
$\max_{\mu \in \mathbb{L}} \mathbb{A} ^\alpha$	5	13	29	49
Min wasted energy	1.35 mJ	10.80 mJ	44.55 mJ	122.40 mJ
Max wasted energy	2.25 mJ	23.40 mJ	117.45 mJ	352.80 mJ

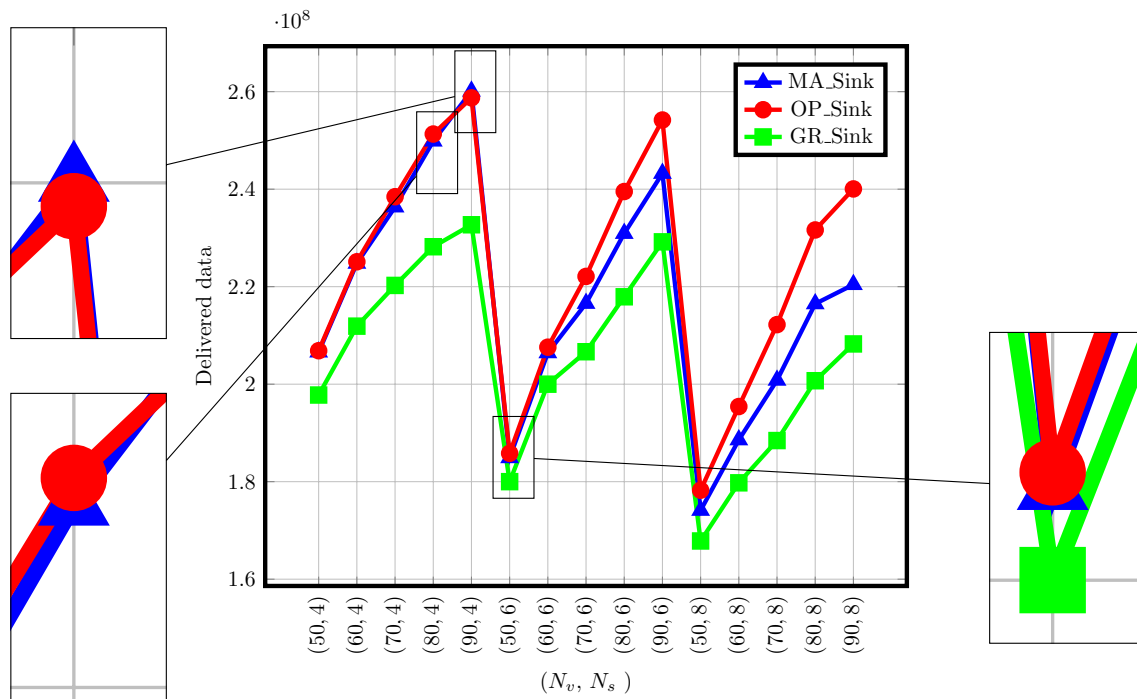


Fig. 7 The effect of changing N_v and N_s on the total delivered data, where $TT_{\max} = 100$ m and $\chi = 10$

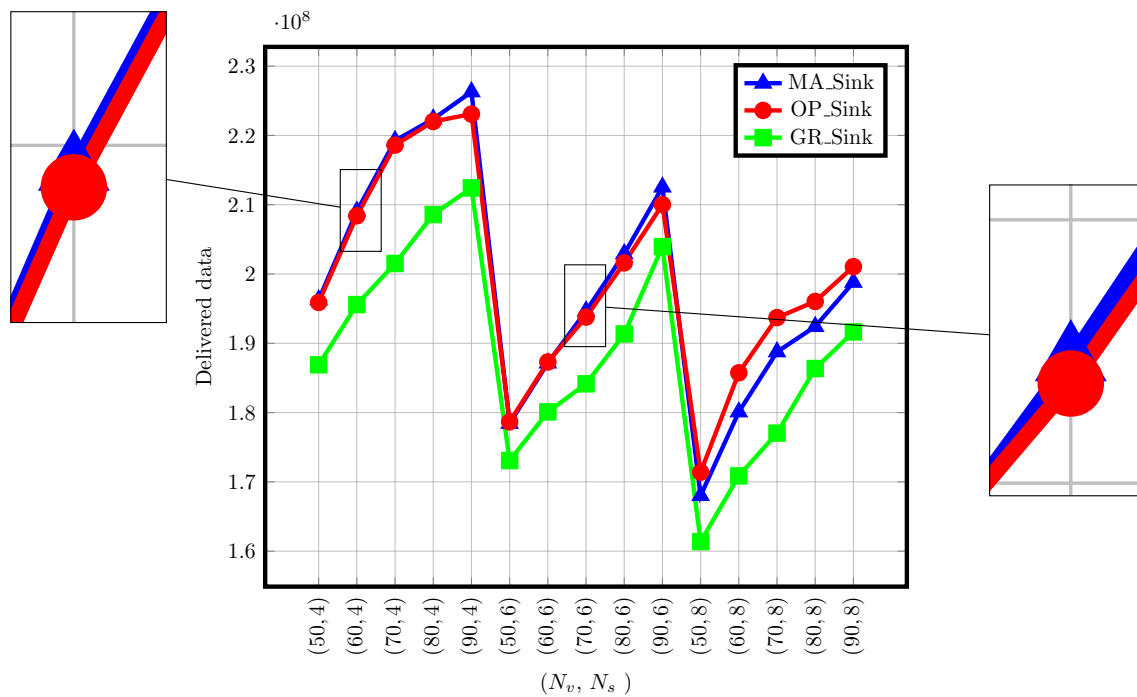


Fig. 8 The effect of changing N_v and N_s on the total delivered data, where $TT_{\max} = 100$ m and $\chi = 20$

Table 8 The volume of χ which MA_Sink overtakes OP_Sink, where $TT_{\max} = 100$ m

		N_v					
		N_s	50	60	70	80	90
$\mathcal{F}_1$	4	25	20	18	15	10	
	6	28	24	20	18	15	
	8	32	29	25	21	18	
$\mathcal{F}_2$	4	30	24	22	19	12	
	6	34	29	24	21	18	
	8	39	35	30	25	22	
$\mathcal{F}_3$	4	27	22	20	16	11	
	6	30	26	22	20	16	
	8	35	32	28	23	20	
$\mathcal{F}_4$	4	22	18	16	13	9	
	6	25	21	18	16	13	
	8	29	26	22	19	16	
$\mathcal{F}_5$	4	20	16	14	12	8	
	6	22	19	16	14	12	
	8	25	23	20	16	14	
$\mathcal{F}_6$	4	19	14	12	10	7	
	6	20	17	14	12	11	
	8	23	20	18	14	13	
$\mathcal{F}_7$	2	17	12	11	9	7	
	3	18	15	12	11	10	
	4	20	18	16	12	12	

In terms of all considered metrics, an increase in the coefficient χ leads to degrading the performance of the methods. In other words, randomness in the traffic generation leads to performance degradation of the methods, especially for PR_Sink where no attribute is of concern in its planning as well as for OP_Sink where the average of attributes is considered. The simulation results indicate that MA_Sink adapts to the heterogeneous traffic generation and the sink can even get more performance with lower complexity. The MA_Sink outperforms OP_Sink in higher heterogeneity due to the two facts, that,

- OP_Sink provides an optimal solution based on the parameters' average, leading to high degradation in its performance
- MA_Sink does not need prior knowledge on when and where the sensors produce data, making performance less prone to heterogeneity.

The volume of χ at which MA_Sink overtakes OP_Sink at different sensor and site counts where $TT_{\max} = 100$ m is tabulated in Table 8. As observed, an increase in the sensor count allows MA_Sink to overtake OP_Sink at a lower χ . This is due to the fact that the diversity among the generated information grows if the sensor count increases. Note that the diversity among the attributes' volumes in the adjacent sites leads to more advantages for applying MA_Sink

because it considers local information about the sites. As the site count increase, the proportionality of the adjacent sites correspond to all sites decreases, leading to a lower performance for MA_Sink, so MA_Sink outperforms OP_Sink at higher χ .

6 Conclusion

This paper considers itinerary planning as a multi-criteria problem, then proposes a multi-objective optimization model and a multi-attribute algorithm. The conflicting criteria: the data merit, data deadline, used memory, latency, travel time, density, and fairness are of concern in this article. In particular, the fuzzy TOPSIS is applied in this article to devise a distributed algorithm for planning an itinerary for the mobile sink. The achievable throughput of the proposed methods is analyzed with respect to varying sensors and site counts and different flow generation types. The simulation results indicate that applying the local information for the proposed algorithm leads to appropriate scalability with minimal reduction in performance. This proposed algorithm is more appropriate to be applied in dynamic and large-scale WSNs with a great count of sensors due to its low complexity. In higher traffic load and heterogeneous WSNs, this algorithm will be more beneficial in comparison with its available counterparts. As an interesting topic for research in the future, developing a multi-attribute trajectory for UAV to collect data from a heterogeneous WSN is of concern.

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